

Quartus II 사용법

VHDL

2009.12.23

Quartus II

설치방법

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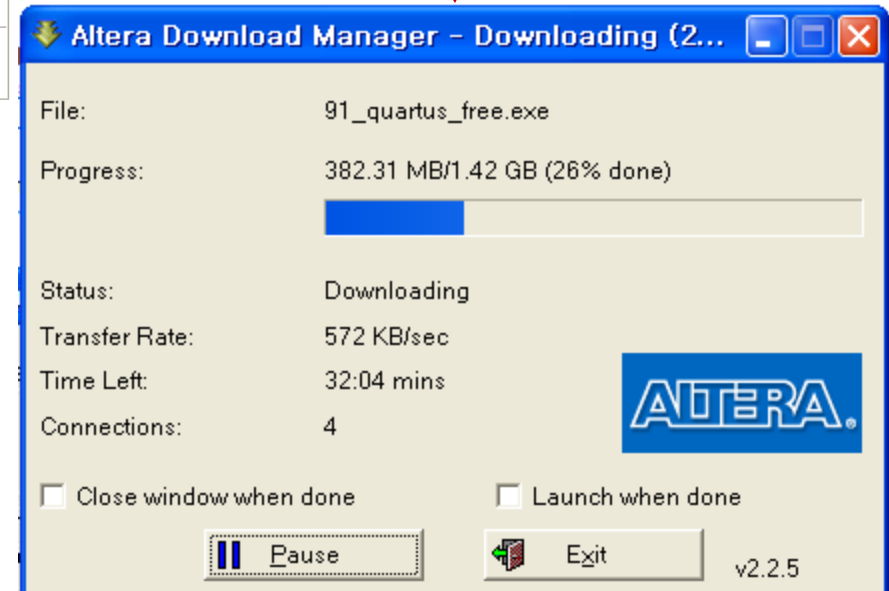
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ModelSim®-Altera® Starter Edition v6.5b for Quartus II v9.1 Red Hat Linux Enterprise 4/5 (32 bits) and SUSE Linux Enterprise 9/10 (32 bits)	Download ▶ No license required	578 MB

Note:

1. To use the Nios II Embedded Design Suite (EDS), you must first install Quartus II software.



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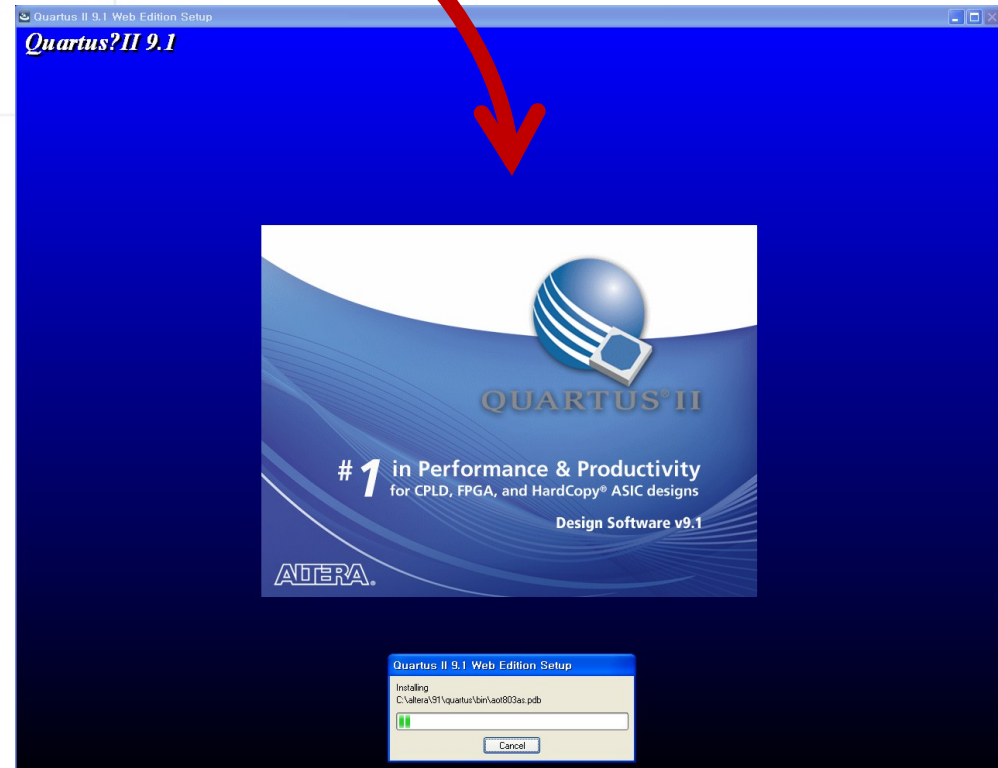
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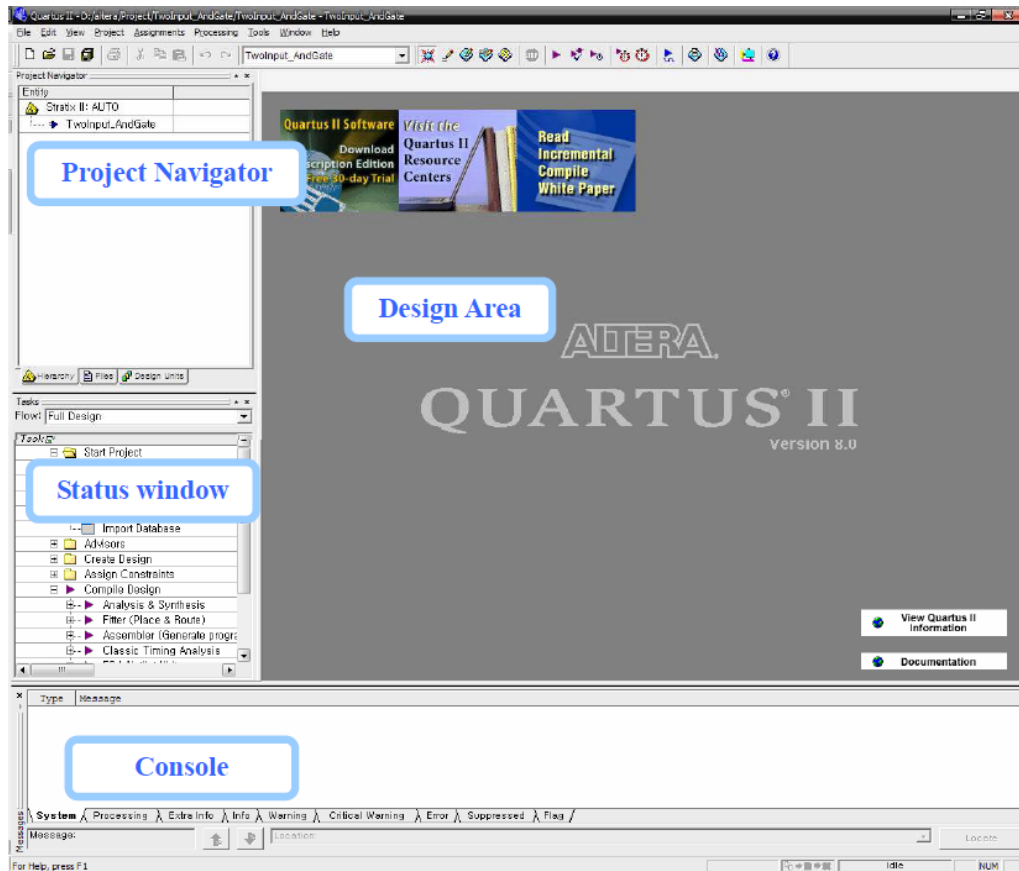
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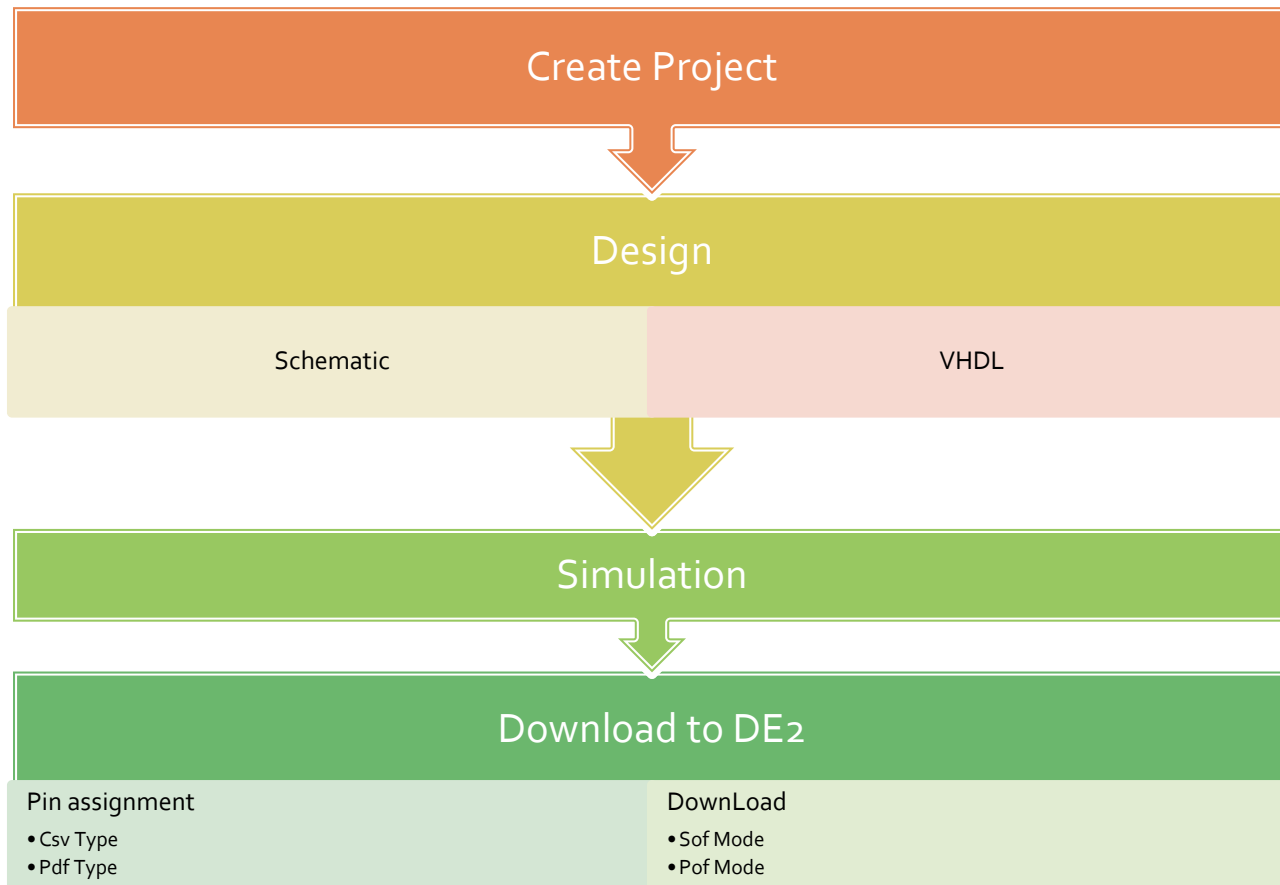


Quartus II의 구성



- Quartus II의 구성은 네 가지 부분으로 되어 있는 것을 볼 수 있다.
- 프로젝트를 생성하였고 이제 Schematic Tool 또는 Verilog를 사용해서 AndGate를 만들어 보자.
- 먼저 메뉴에서 File-New를 클릭하자!!

Quartus II의 사용 단계



Create Project

Create Project

New Project Wizard: Directory, Name, Top-Level Entity [page 1 of 5]

What is the working directory for this project?

C:\altera\91\quartus\MyProject\ch2

What is the name of this project?

test

What is the name of the top-level design entity for this project? This name is case sensitive and must exactly match the entity name in the design file.

test

Use Existing Project Settings ...

< Back Next > Finish 취소

1. 다이얼로그 창이 생성될 것인데 다음과 같은 순서로 프로젝트를 생성하자
2. 간단한 설명이 나오는데 그냥 Next를 클릭
3. 프로젝트 경로를 설정하며 밑의 그림의 경우
C:\altera\Project\TwoInput_AndGate 디렉토리를 생성하고 프로젝트 이름도 TwoInput_AndGate로 하였다.
4. 기존의 디자인 했던 파일이 있다면 Import 시킬 수 있다. 지금은 새롭게 출발하는 과정이니 Next만 클릭한다.

Create Project_타겟 디바이스 설정

New Project Wizard: Family & Device Settings [page 3 of 5]

Select the family and device you want to target for compilation.

Device family
Family: Cyclone II
Devices: All

Target device
☐ Auto device selected by the Filter
☒ Specific device selected in 'Available devices' list

Show in 'Available device' list
Package: Any
Pin count: Any
Speed grade: Any
☒ Show advanced devices
☐ HardCopy compatible only

Available devices:

Name	Core v...	LEs	User I/...	Memor...	Embed...	PLL	Global ...
EP2C35F484C7	1.2V	33216	322	483840	70	4	16
EP2C35F484C8	1.2V	33216	322	483840	70	4	16
EP2C35F48416	1.2V	33216	322	483840	70	4	16
EP2C35F672C6	1.2V	33216	475	483840	70	4	16
EP2C35F672C7	1.2V	33216	475	483840	70	4	16
EP2C35F672C8	1.2V	33216	475	483840	70	4	16
EP2C35F67218	1.2V	33216	475	483840	70	4	16
EP2C35U484C6	1.2V	33216	322	483840	70	4	16
EP2C35U484C7	1.2V	33216	322	483840	70	4	16

Companion device
HardCopy:
☒ Limit DSP & RAM to HardCopy device resources

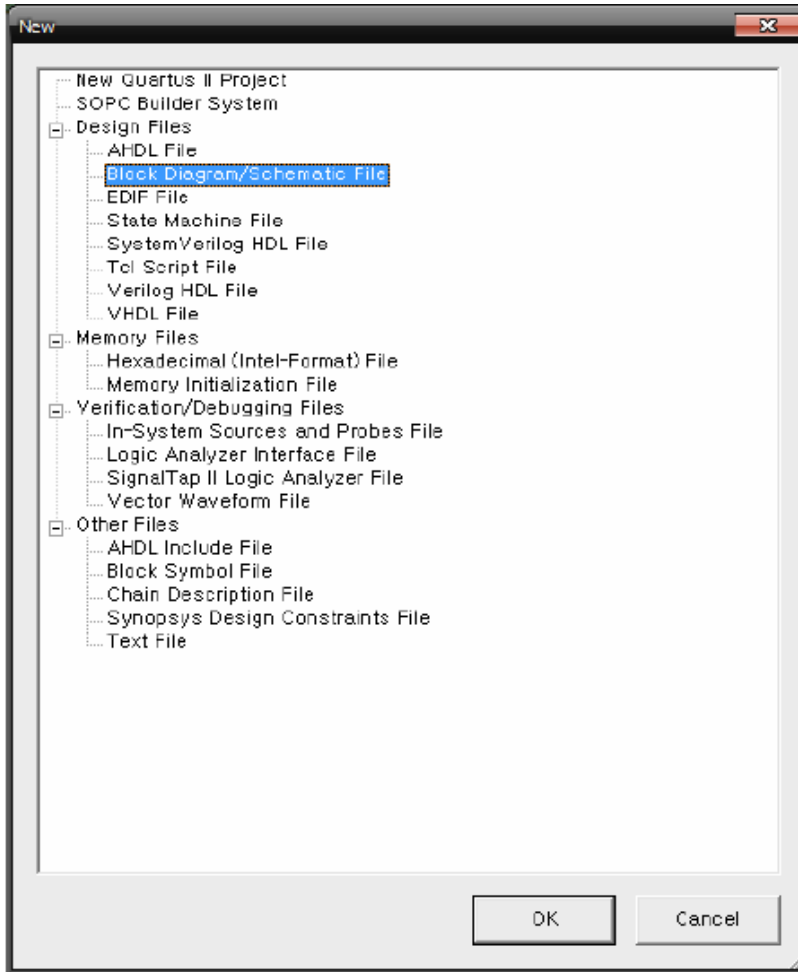
< Back Next > Finish 취소

Device Family : Cyclone II
Available Devices: EP2C35F672C6

1. 알테라 테스트 보드를 통해서 설계한 회로를 테스트 할 수 있으며 목록들은 보드와 칩셋을 나열한 것이다.
2. 특정한 컴파일러나 모델을 사용할 때 사용하며 특이한 것이 필요 없을 때에는 모두 None 으로 설정한다.
3. 마지막 다이얼로그는 프로젝트를 설정 했을 때의 요약정보를 보여준다. Finish를 눌러서 프로젝트 설정을 마무리 하자!

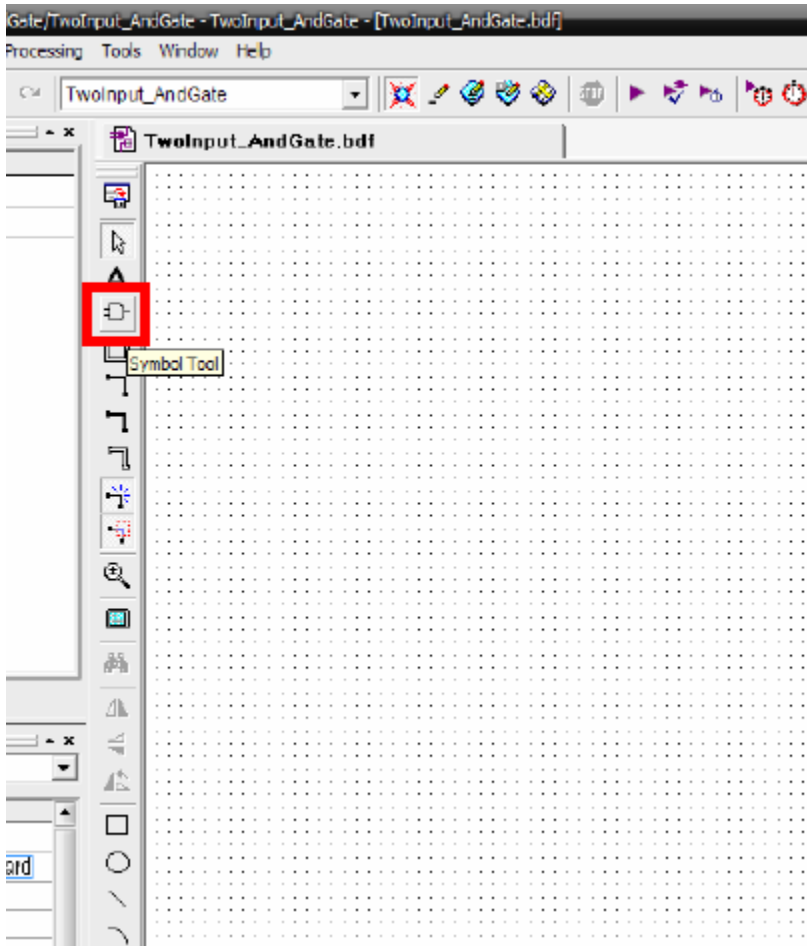
Schematic Design

Schematic Design



1. 먼저 Block Diagram/Schematic File 을 만들어 보도록 하자!!
2. 그럼 Design Area에 Block1.bdf가 생성됐을 것이다.
먼저 파일이름을 프로젝트 이름과 일치 시킨다.
3. 이번 예제에서는 TwoInput_AndGate.bdf 새 이름으로 저장한다.

Schematic Design



이제 AndGate를 그릴 준비가 되었다.
입력 두개의 AndGate를 생각해 보면 입력이 두개 필요하고 출력이 하나 필요할 것이다. 그리고 중요한 AndGate가 필요할 것이다.

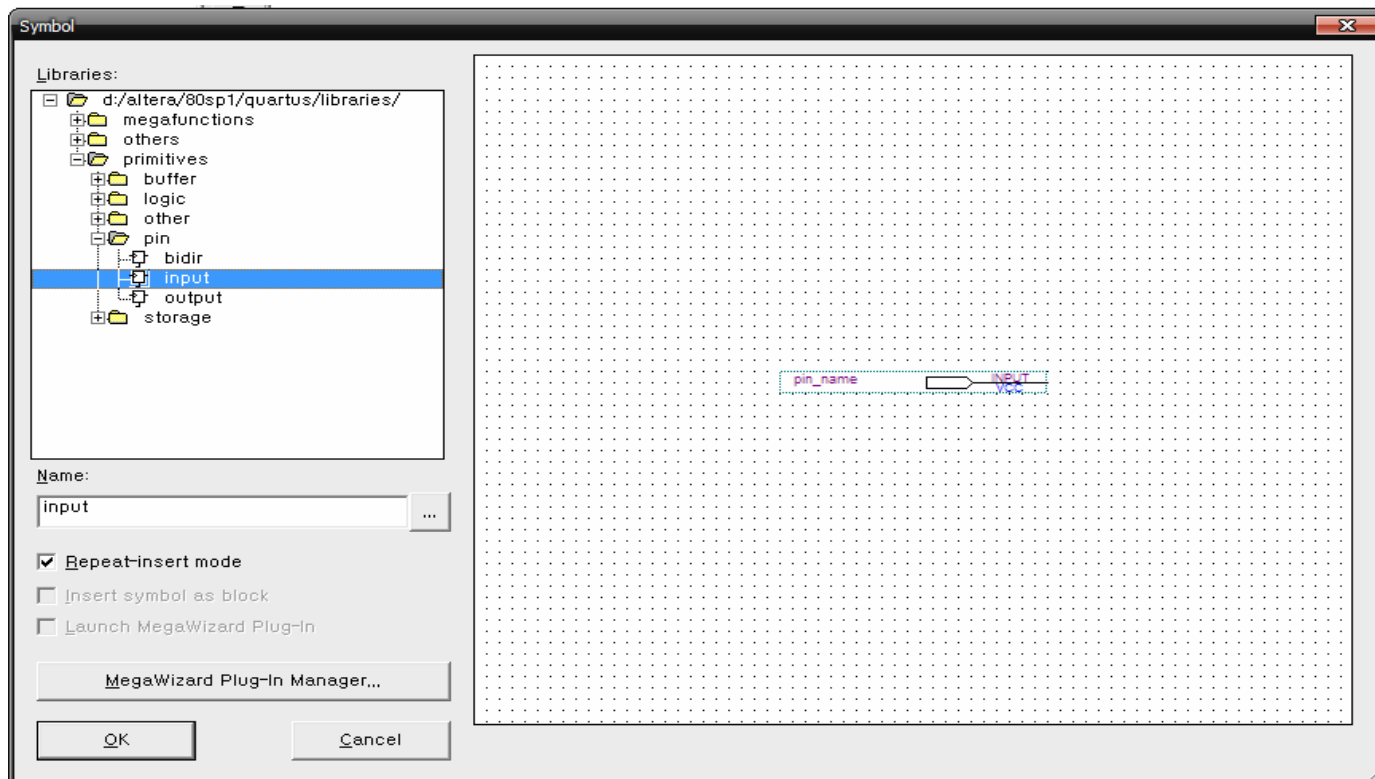
입력 값과 출력 값에 이름을 준다면 입력에는 InputA, InputB 출력에는 OutputC로 정의할 수 있을 것이다.

1. Symbol 툴을 클릭하자.

Schematic Design

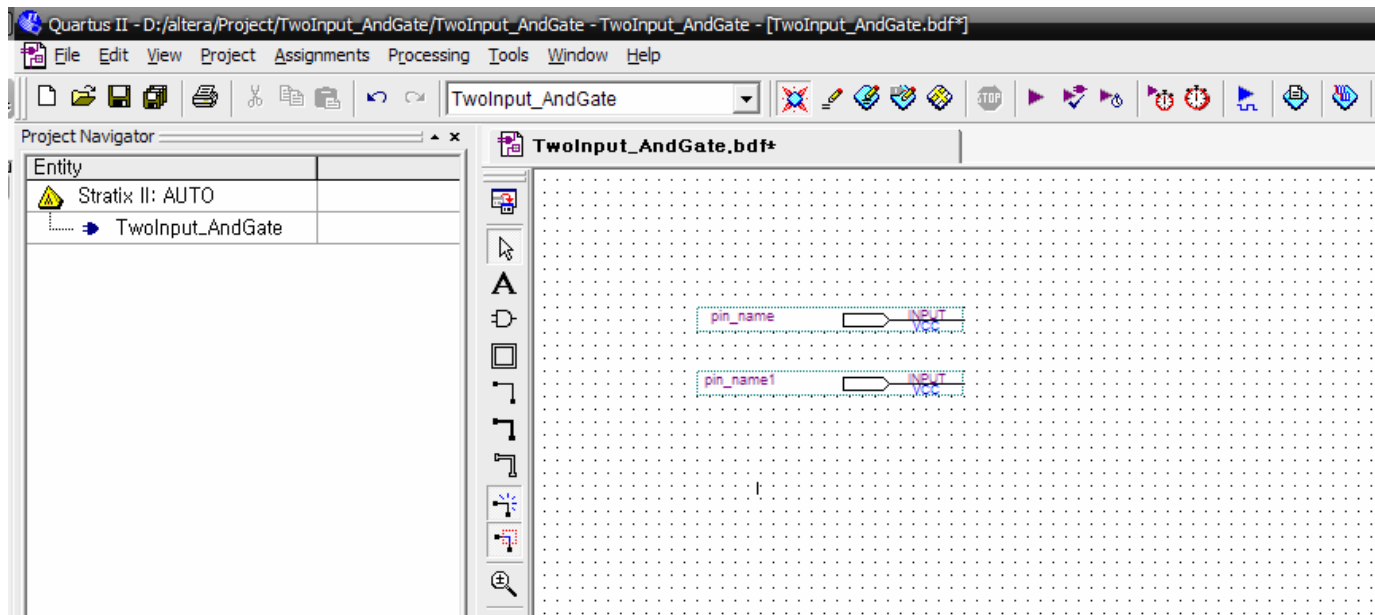
2. Name Text Box 에서 input이라고 입력을 하는 방법과 직접 메뉴에서 찾는 방법이 있다.

메뉴에서 symbol을 찾을 때에는 primitives/pin/input 경로에서 찾을 수 있다. OK를 누르자



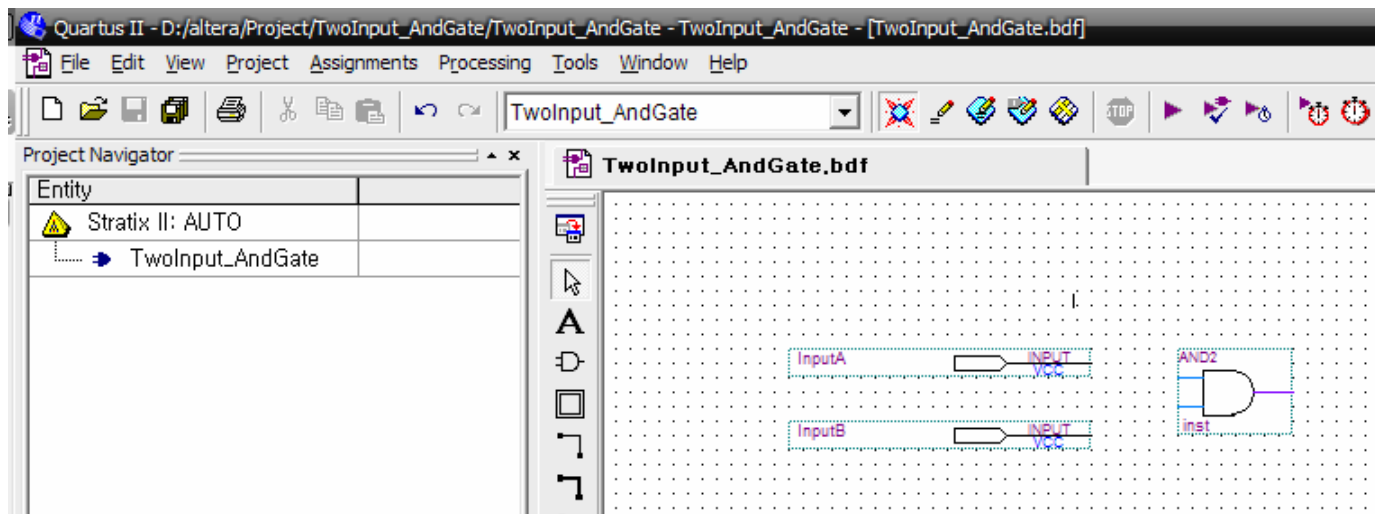
Schematic Design

- Input Pin 생성



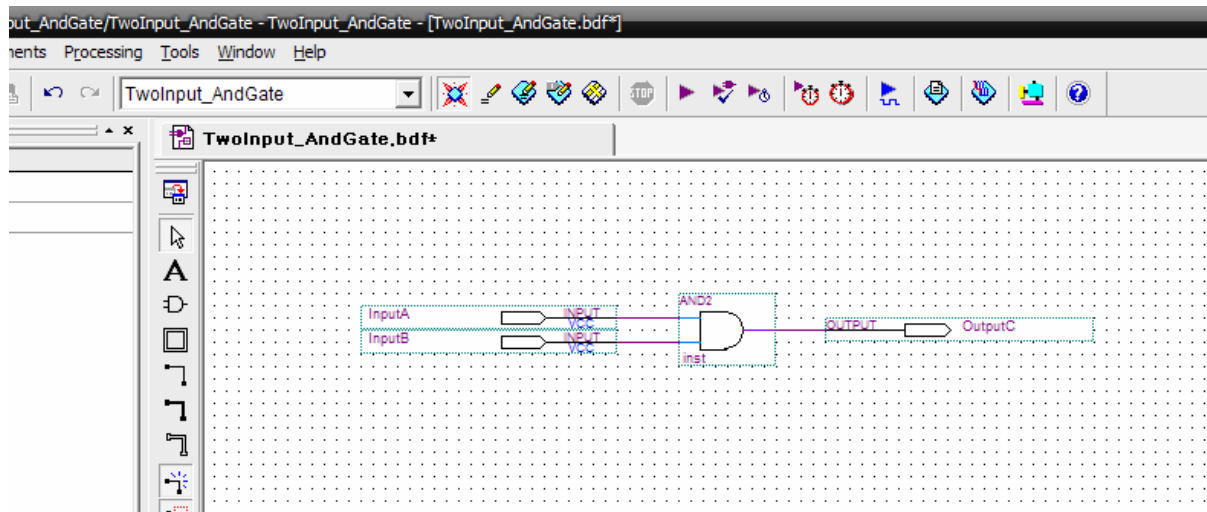
Schematic Design

3. 다음 순서는 symbol 창에서 AndGate를 찾아 보자.
Input Pin가 마찬가지로 And2를 입력을 하던가
primitives/logic/and2 경로에서 symbol을 찾을 수 있다.



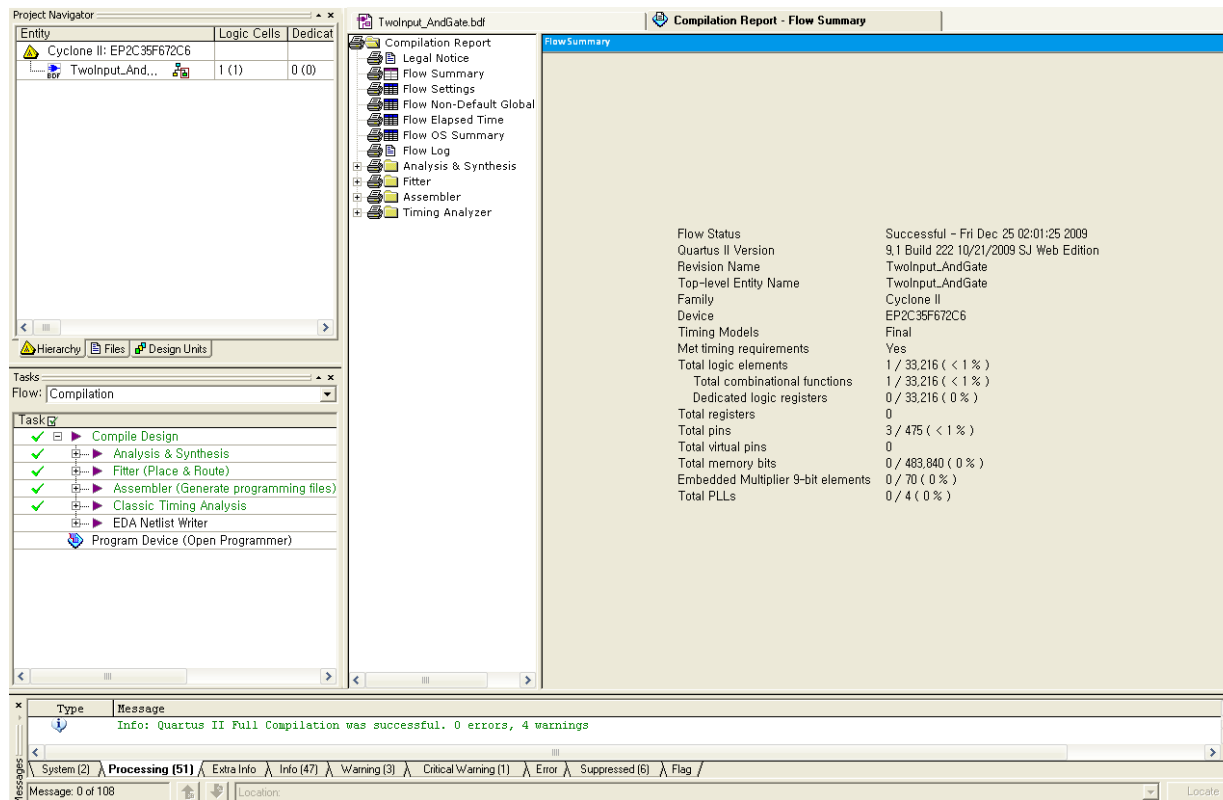
Schematic Design

4. Input Pin과 And Gate를 연결하여 보자!!
5. 다음 순서로 symbol 창에서 Output을 찾고 AndGate와 연결하자. 이로서 AndGate 회로가 완성되었다.



Schematic Design_Compile

- 여기서 끝나는 것이 아니라 컴파일 과정과 시뮬레이션 과정이 남아 있다. 일단 컴파일을 통해서 회로가 잘 연결되어 있는지 확인해 보자.
- Ctl+L 를 눌러서 컴파일 하자. 에러가 없다면 그림 15과 같은 그림이 나타날 것이다.
- Console 에서 Info: Quartus II Full compilation was successful. 0 errors, 6 warnings 같은 메시지가 나타났을 것이다.



VHDL Design

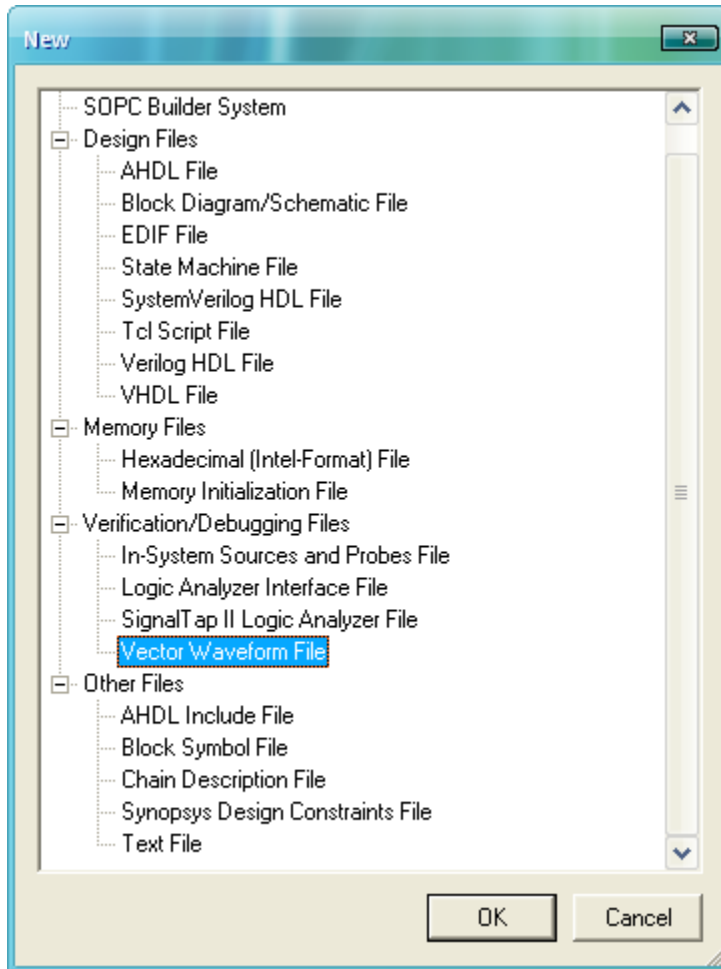
VHDL Design

```
1  Library ieee;
2  Use ieee.std_logic_1164.ALL;
3
4  entity TwoInput_AndGate is
5  port (InputA, InputB      : in bit;
6        OutputC            : out bit);
7  end TwoInput_AndGate;
8
9  architecture And_logic1 of TwoInput_AndGate is
10 begin
11     OutputC <= InputA and InputB;
12 end And_logic1;
13
```

- Project를 만들 때의 Entity name과 VHDL 소스 안의 Entity name을 일치시켜줘야 한다.

Simulation

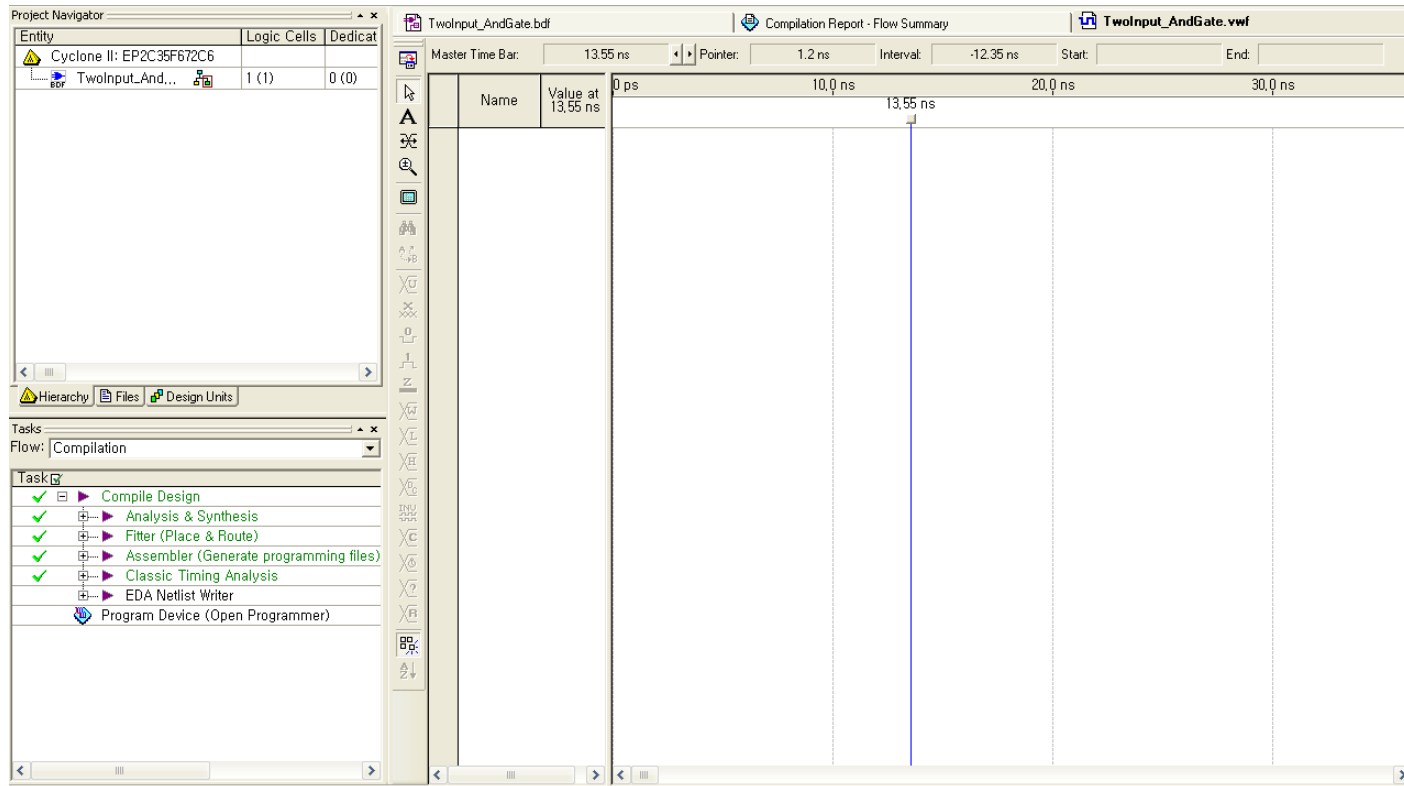
Simulation



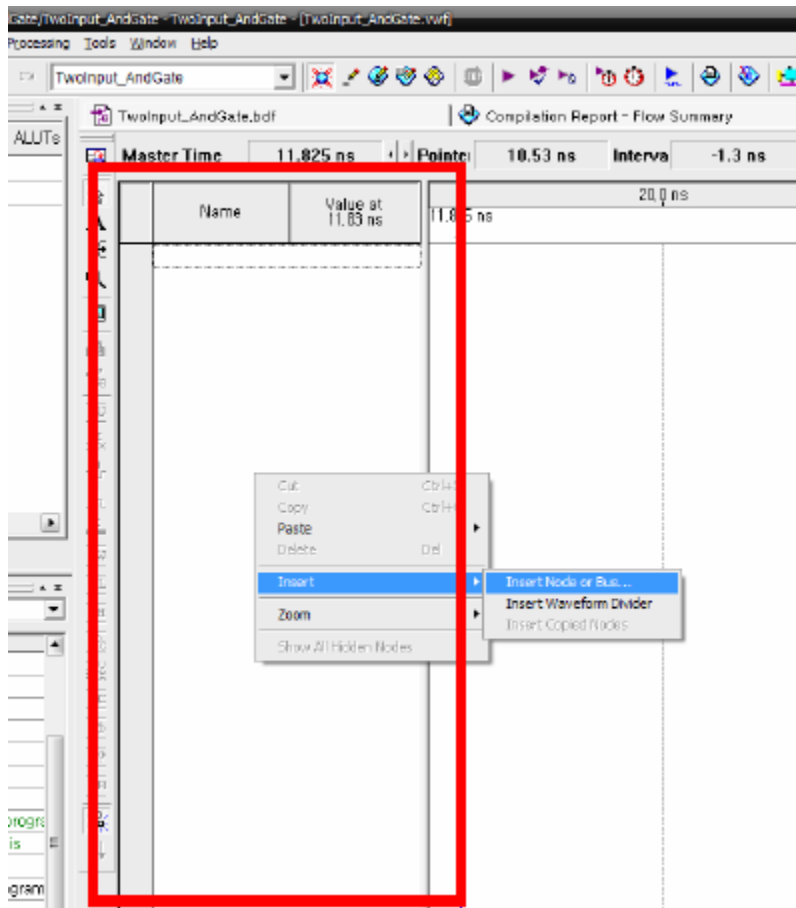
- And Gate 회로의 로직이 잘 동작하는지 확인하는 일이 남아 있다. 확인하는 작업은 Vector Waveform file을 생성해 보자.
- 그리고 vwf 파일 이름을 프로젝트 이름과 동일하게 변경하여야 한다.

Simulation

■ Waveform Layout

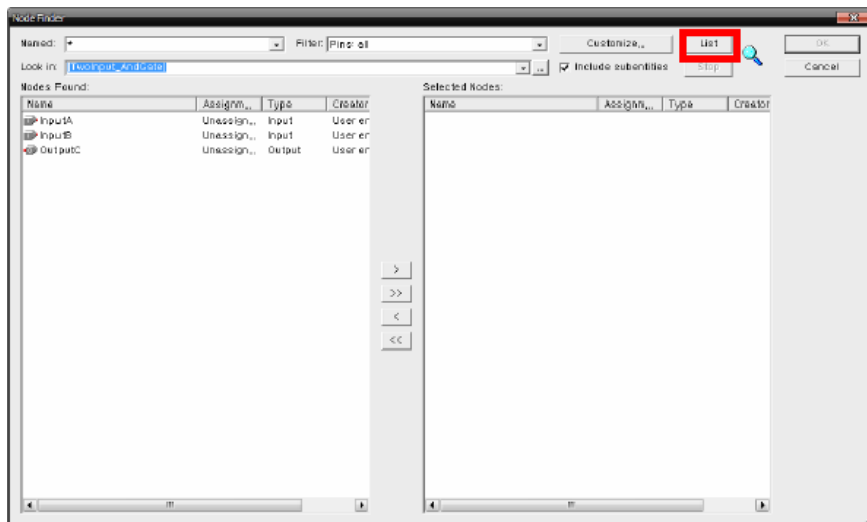
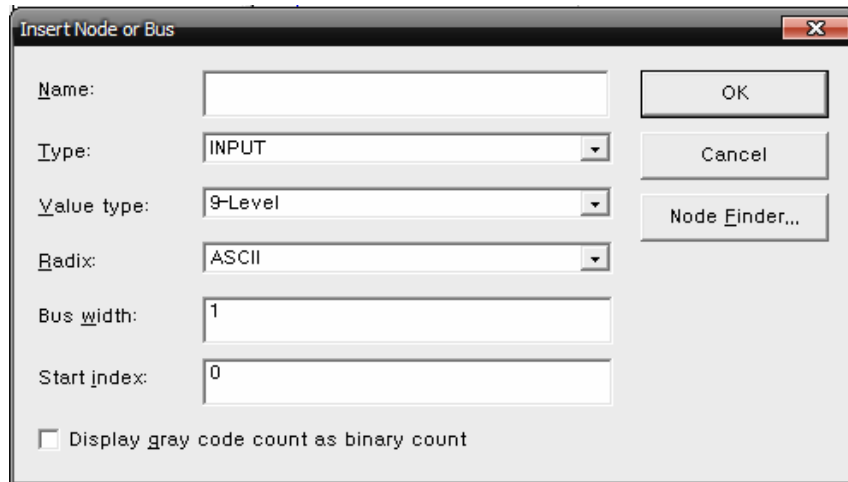


Simulation



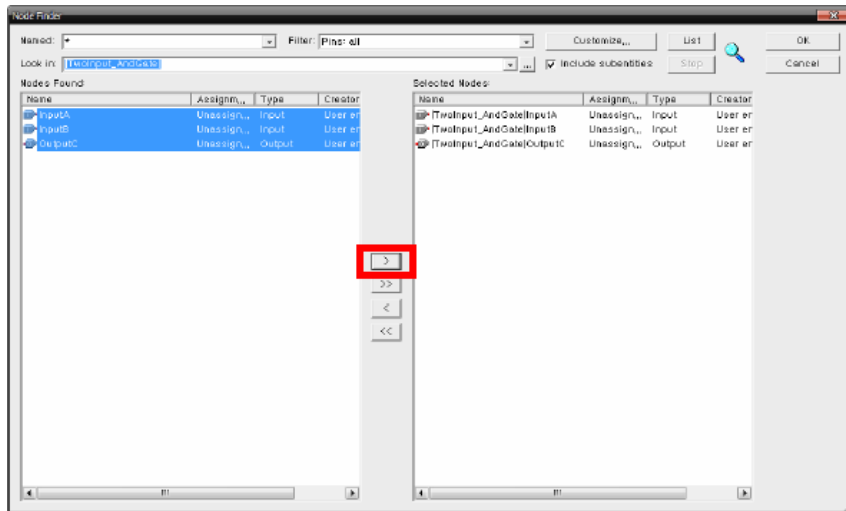
- Red Box 부분을 더블 클릭하거나 팝업 창에서 Insert Node를 선택한다

Simulation

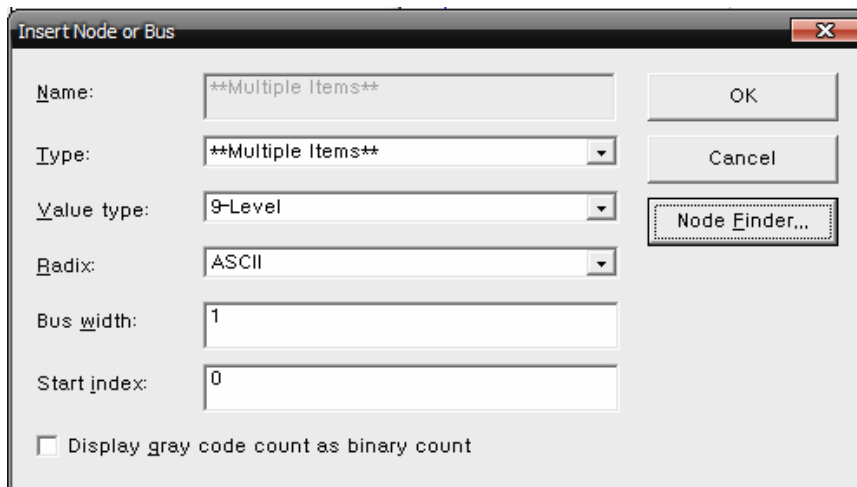


- Node Finder 버튼을 클릭한다.
- List Button을 클릭하면 BDF에서 생성된 입력, 출력 핀이 검색된다.

Simulation

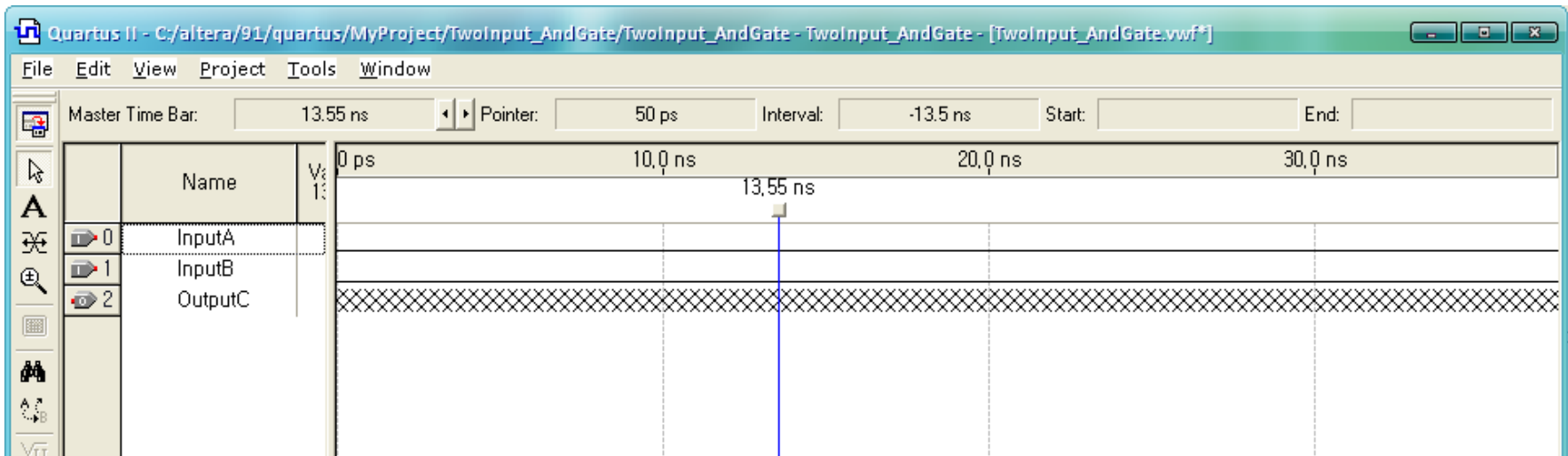


- 시뮬레이션 할 Pin을 선택하고 화살표를 클릭한다. 그런 다음 OK 버튼을 누른다.
- OK 버튼을 누른다.

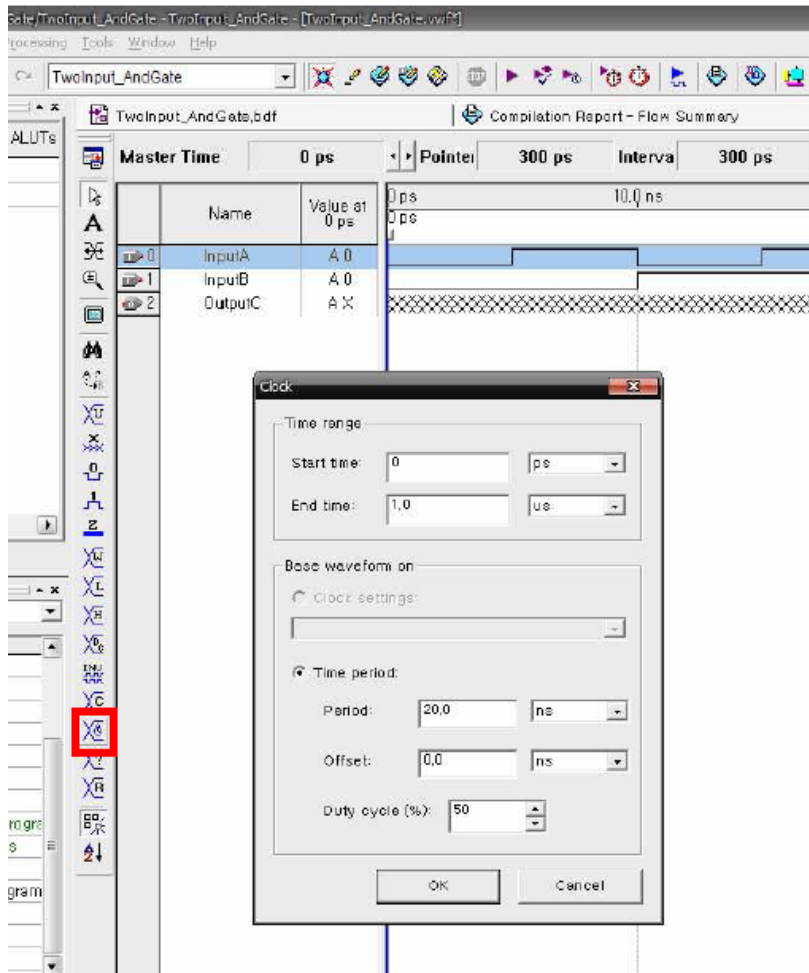


Simulation

- Input과 Output 핀이 나타나는 것을 확인할 수 있다.



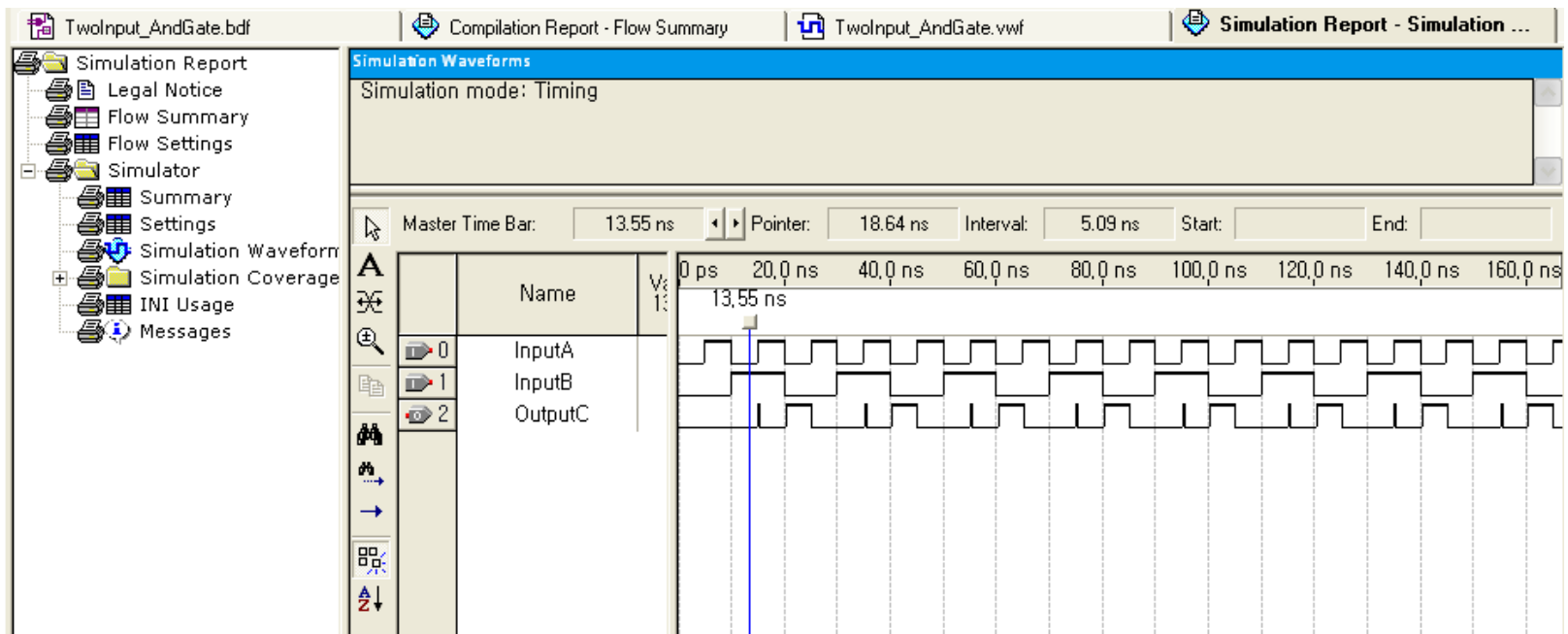
Simulation



- 컨트롤 하려는 핀을 선택한다.
- Overwrite Clock을 선택하고 Period을 조정한다.

Simulation_Compile

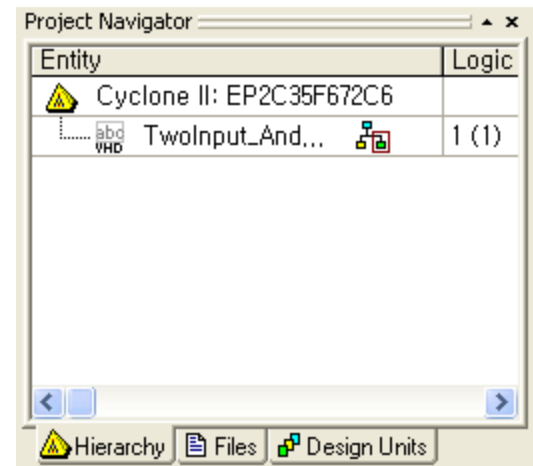
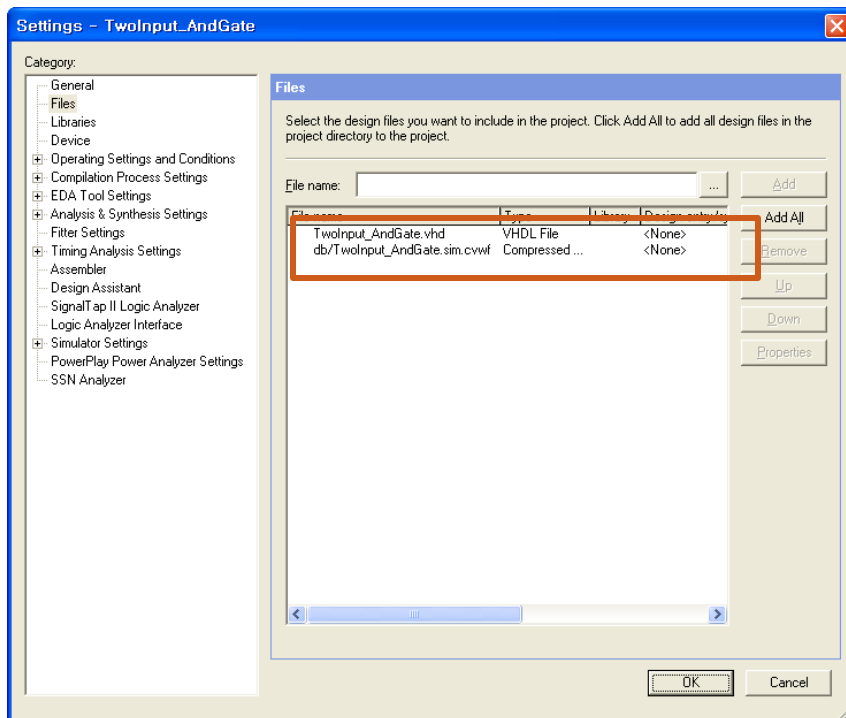
- Ctrl+I를 누르거나 메뉴 [processing – start simulation] 눌러 simulation을 실행한다.
- Simulation이 성공적으로 실행되면 OutputC 핀의 Value가 기록된다. 유의할 점은 **Gate**마다 딜레이가 있기 때문에 결과값이 밀려 보이는 것이 이상한 것이 아니다.



Download to DE2

Add Files in Project

- Project -> Add Files in Project -> 현재 VHDL 파일을 Project에 포함시킨다.



Pin Assignment

- Assignments -> Assignment Editor -> Pin Category

The screenshot shows the 'Assignment Editor' window. The 'Category' dropdown is set to 'Pin'. The 'Edit' field shows '<<new>>'. Below is a table with columns: To, Location, I/O Bank, I/O Standard, General Function, Special Function, and Reserved. The first row shows '1' in the 'To' column and '<<new>>' in the 'Location' column.

	To	Location	I/O Bank	I/O Standard	General Function	Special Function	Reserved
1	<<new>>	<<new>>					

Pin Assignment

- Pin번호를 할당하고 저장한다.
- 다시 컴파일을 수행한다.

TwoInput_AndGate.vhd | TwoInput_AndGate.vwf | Simulation Report - Simulation Waveforms | Assignment Editor

Category: Pin

Information: This cell specifies the pin name to which you want to make an assignment. --> Double-click to create a new assignment.

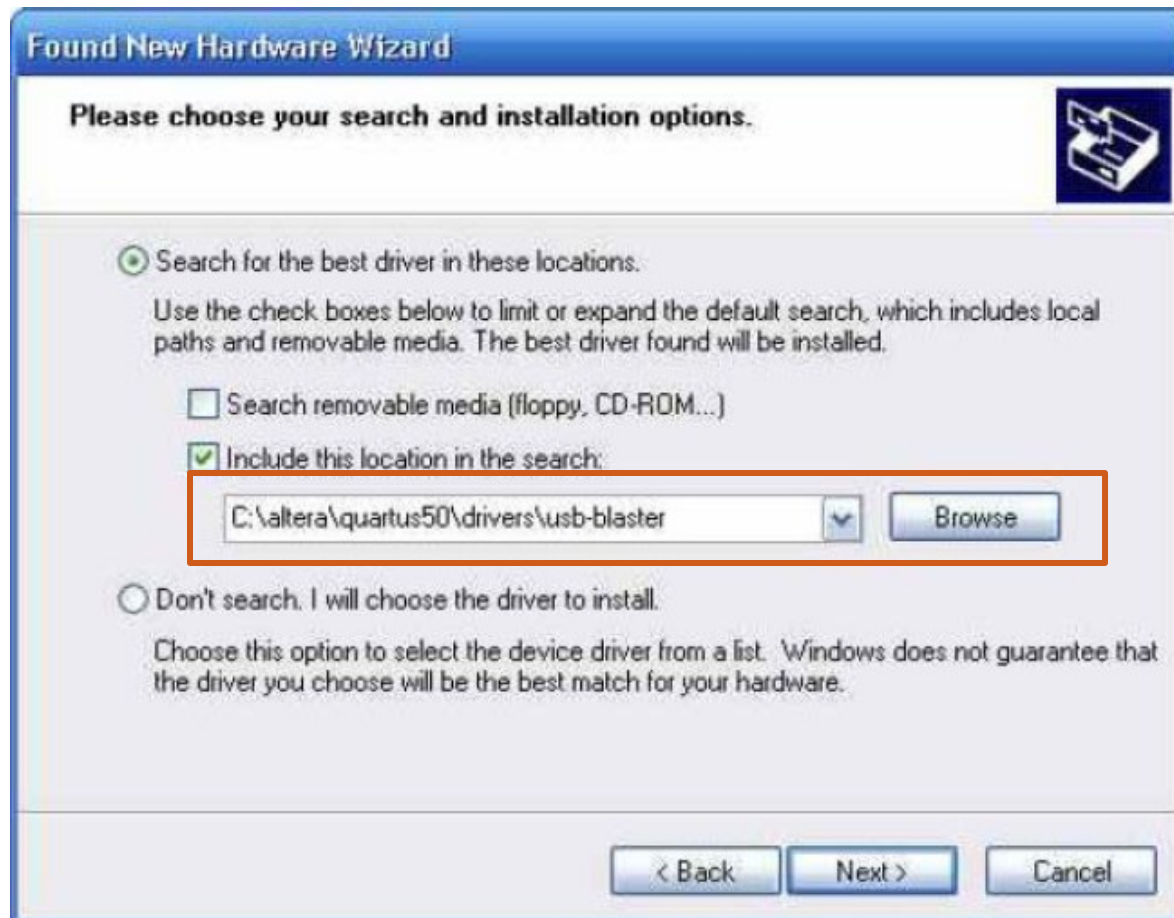
Edit: <<new>>

	To	Location	I/O Bank	I/O Standard	General Function	Special Function	Reserved	Enabled
1	InputA	PIN_N25	5	3.3-V LVTTL	Dedicated Clock	CLK4, LVDSCLK2p, In...		Yes
2	InputB	PIN_N26	5	3.3-V LVTTL	Dedicated Clock	CLK5, LVDSCLK2n, In...		Yes
3	OutputC	PIN_AE23	7	3.3-V LVTTL	Column I/O	LVDS151n		Yes
4	<<new>>	<<new>>						

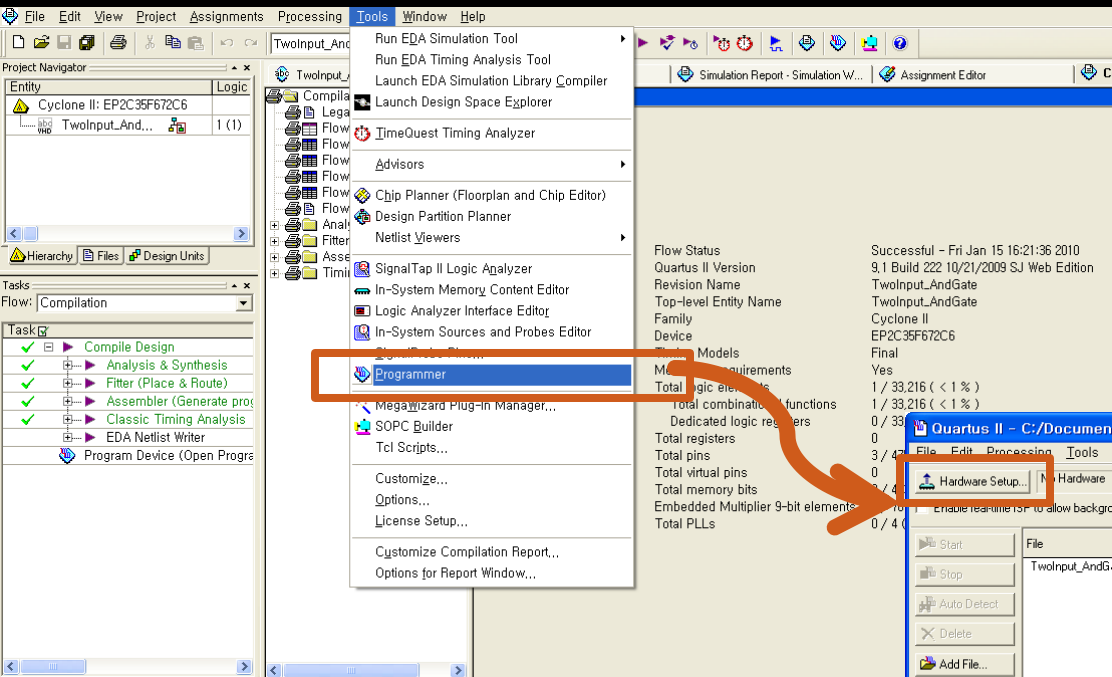
Compile _ Warning 3, Error 0

Info: Fitter is performing an Auto Fit compilation, which may decrease Fitter effort to reduce compilation time
Warning: Feature LogicLock is only available with a valid subscription license. Please purchase a software subscription to gain full access to this feature.
Info: Device migration not selected. If you intend to use device migration later, you may need to change the pin assignments as they may be incompatible with the current configuration.
Info: Fitter converted 3 user pins into dedicated programming pins
Info: Timing-driven compilation is using the Classic Timing Analyzer
Info: Timing requirements not specified -- quality metrics such as performance and power consumption may be sacrificed to reduce compilation time.
Info: Starting register packing
Info: Finished register packing
Info: Fitter preparation operations ending: elapsed time is 00:00:01
Info: Fitter placement preparation operations beginning
Info: Fitter placement preparation operations ending: elapsed time is 00:00:00
Info: Fitter placement operations beginning
Info: Fitter placement was successful
Info: Fitter placement operations ending: elapsed time is 00:00:02
Info: Fitter routing operations beginning
Info: Average interconnect usage is 0% of the available device resources
Info: Fitter routing operations ending: elapsed time is 00:00:01
Info: The Fitter performed an Auto Fit compilation. Optimizations were skipped to reduce compilation time.
Info: Started post-fitting delay annotation
Warning: Found 1 output pins without output pin load capacitance assignment
Info: Delay annotation completed successfully
Warning: The Reserve All Unused Pins setting has not been specified, and will default to 'As output driving ground'.

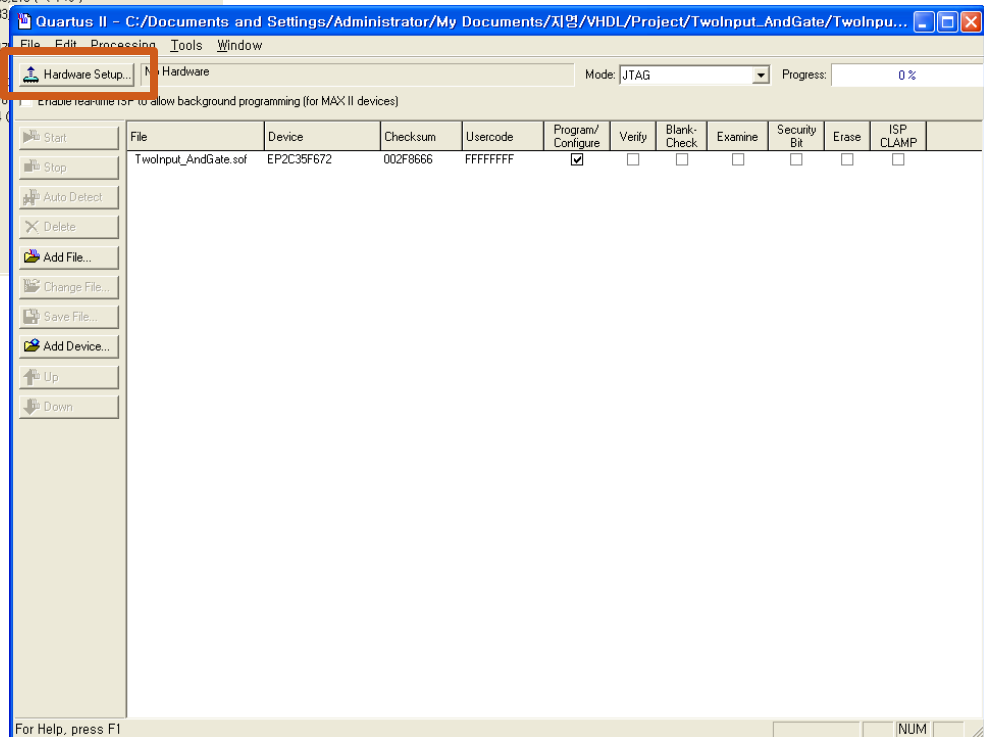
USB-Blaster Driver 설치



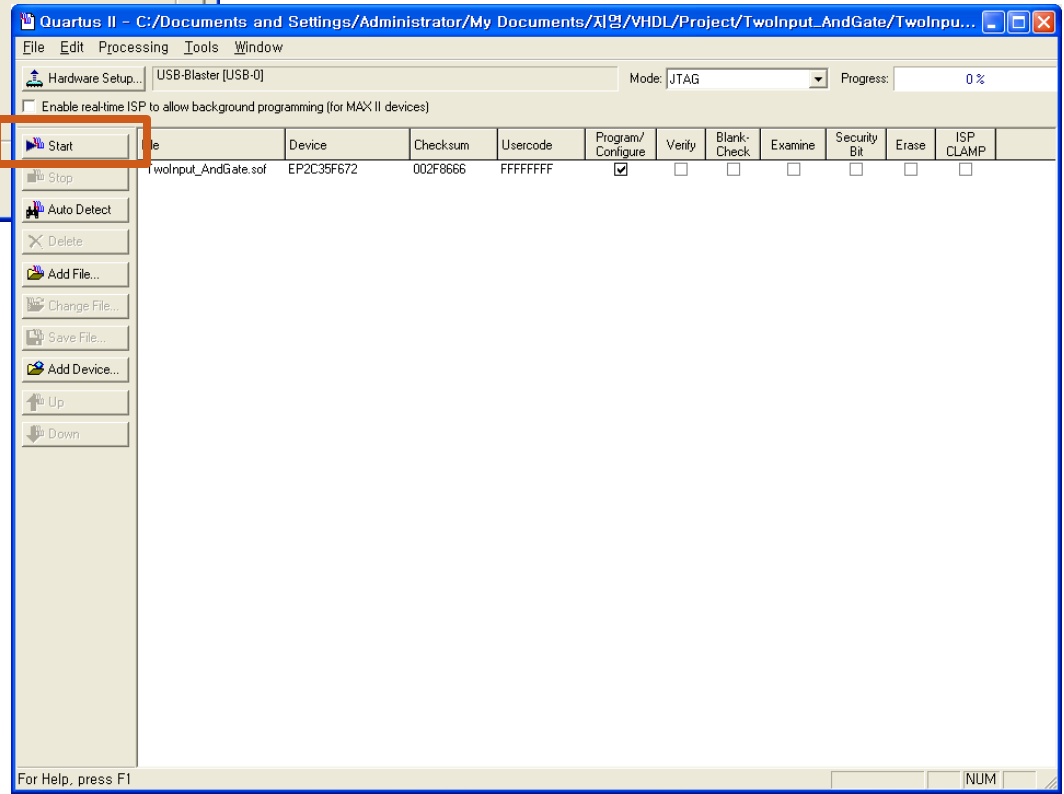
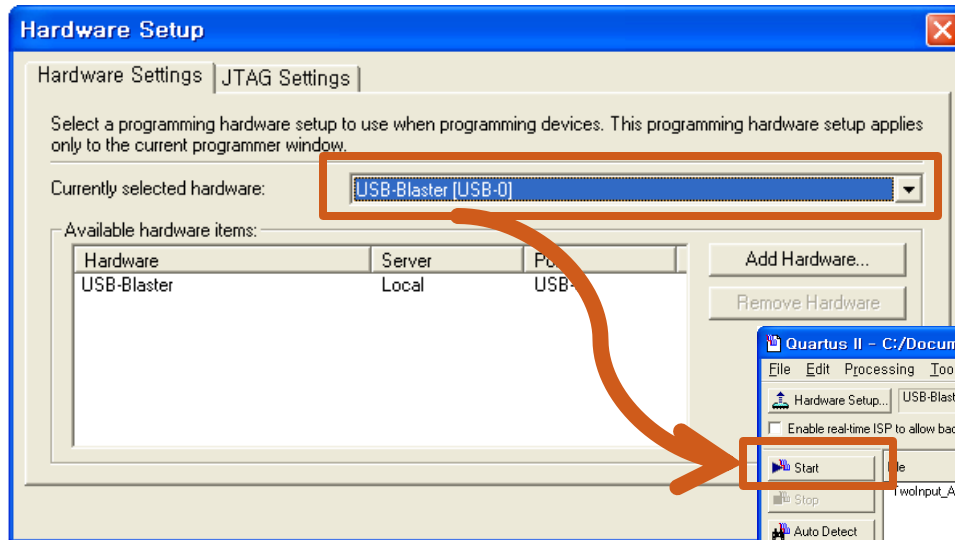
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Tools -> Programmer ->
"TwoInput_AndGate.sof"파일을 다운로드 하고 DE2로 동작을 확인한다.



Download to DE2



참고자료

- <http://chorogyi.tistory.com/attachment/ik240000000000.pdf>
- Pin Assignment
http://instruct1.cit.cornell.edu/courses/ece576/DE2/DE2_pin_assignments.csv
http://www.terasic.com.tw/attachment/archive/30/DE2_Pin_Table.pdf